

Differentiable Inverse DC-OPF: Recovering Generator Costs and Capacities from Dispatch Data

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Abstract—We study the inverse DC optimal power flow (DC-OPF) problem: given a sequence of demand–dispatch pairs $\{(d_t, g_t)\}_{t=1}^T$ produced by an unknown linear merit-order DC-OPF, recover the per-generator marginal costs f^* , generator capacities $g_{\max}^*$, and line capacities $p_{\max}^*$. We pose this as a constrained convex regression problem and solve it end-to-end by differentiating through a CVXPY-Layer that embeds the convex forward QP. The resulting estimator generalises the classical KKT-residual baseline of Keshavarz, Wang, and Boyd [1] and the implicit-differentiation estimator of Fuentes-Valenzuela and Degleris [3] by jointly estimating costs *and* capacities, supporting Laplacian-stratified time-of-day costs, and yielding exact downstream sensitivities (e.g. marginal emission factors). On synthetic networks our estimator attains cosine recovery 0.96 for the cost vector versus 0.91 for the KKT baseline and Spearman 0.92 versus 0.89, with roughly $5\times$ tighter seed variance; on an eight-fuel PJM-like economic-dispatch problem the stratified estimator recovers the marginal merit order to 80% pairwise accuracy. We further give an active-set identifiability lemma and an empirical identifiability score that explains which generators are recoverable from data.

I. INTRODUCTION

Modern power-system operators publish enormous quantities of dispatch data—hourly generation, demand, and clearing prices—without ever disclosing the underlying cost curves or operating limits used by their market clearing engine. Recovering these latent parameters is central to many downstream tasks: forecasting carbon emissions per unit of new load (marginal emission factors, MEFs), counterfactual policy analysis, calibrating capacity-expansion models, and detecting market power. We formulate this recovery as an *inverse* optimization problem [4], [5].

The forward DC-OPF is a small convex QP with linear costs, equality power balance, and box bounds on generation and line flow. Two main families of estimators exist. (1) The KKT-residual approach of Keshavarz, Wang, and Boyd [1]: assume an active set, and solve a single convex QP that minimises the squared norm of the stationarity residual over the cost vector and dual multipliers. (2) The implicit-differentiation approach pioneered by Agrawal et al. in CVXPY-Layers [2] and applied to inverse OPF by Fuentes-Valenzuela and Degleris [3]: differentiate through the QP solution map and minimise a forward-prediction loss by gradient descent.

We make four contributions.

- 1) A unified, fully-open implementation that supports both transport and DC physics, jointly estimates $(f, g_{\max}, p_{\max})$, and admits a stratified-by-stratum cost parameterisation with cycle Laplacian smoothing.
- 2) A clean head-to-head comparison against the KKT-residual baseline and pure regression baselines (ridge, MLP) under controlled synthetic noise, with multi-seed mean $\pm$ std error bars.
- 3) An *active-set identifiability score* that empirically ranks per-generator recovery error and a corresponding identifiability lemma.
- 4) Validation on a PJM-like single-bus economic-dispatch problem with eight realistic fuel buckets and time-varying renewable availability, plus a downstream MEF accuracy study.

All code, configurations, and figures are released at <https://github.com/anonymous/ee364b-inverse-opf>.

II. FORWARD AND INVERSE DC-OPF

Let $A \in \mathbb{R}^{n \times m}$ be the bus–line incidence of a connected n -bus, m -line graph, and let $b \in \mathbb{R}_{++}^m$ collect the line susceptances. The forward DC-OPF is

$$\begin{aligned} & \underset{g \in \mathbb{R}^n, p \in \mathbb{R}^m, \theta \in \mathbb{R}^n}{\text{minimize}} && f^\top g + \frac{\varepsilon}{2} \|g\|_2^2 + \frac{\tau}{2} \|p\|_2^2 \\ & \text{subject to} && Ap + g = d, \\ & && p = \text{diag}(b) A^\top \theta, \\ & && 0 \leq g \leq g_{\max}, |p| \leq p_{\max}, \\ & && \theta_{\text{ref}} = 0. \end{aligned} \tag{1}$$

The Tikhonov terms $\varepsilon, \tau > 0$ make the QP strongly convex, so the solution map $(d, f, g_{\max}, p_{\max}) \mapsto (g^*, p^*)$ is unique and Lipschitz. We use $\varepsilon = 10^{-1}, \tau = 10^{-2}$ throughout. Setting $b \equiv 1$ recovers the network-flow / transport relaxation; we use this as our default since it isolates the inverse-cost problem from network sensitivity.

Inverse problem: Given $\{(d_t, g_t^{\text{obs}})\}_{t=1}^T$ with $g_t^{\text{obs}} = g^*(d_t; \theta^*) + \eta_t$, recover $\theta^* = (f^*, g_{\max}^*, p_{\max}^*)$. Several scaling and sign symmetries exist (e.g. rescaling f^* by any positive constant leaves the merit order invariant), so we use scale-invariant recovery metrics: cosine $\cos \angle(\hat{\theta}, \theta^*)$, Spearman ρ , and *merit-order accuracy*—the fraction of generator pairs (i, j) whose ordering $f_i \leq f_j$ is preserved.

III. IDENTIFIABILITY

A generator’s cost is observable only when its dispatch is sensitive to demand changes, which occurs precisely when the generator is strictly interior of its box constraint at the solution. We make this precise.

Lemma 1 (Active-set identifiability). *Fix $(d, g_{\max}, p_{\max})$. If at $g^*(d; f)$ generator i is strictly interior, $0 < g_i^* < g_{\max, i}$, and the KKT system is non-degenerate (strict complementarity), then the partial derivative $\partial g_i^* / \partial f_i$ is non-zero. Conversely, if generator i is at its lower or upper bound for all $t = 1, \dots, T$, then f_i is unidentifiable from $\{(d_t, g_t)\}$ up to translations within $[\min_{j: g_j > 0} f_j, \infty)$ (resp. $[0, \max_{j: g_j < g_{\max, j}} f_j]$).*

The proof is a standard implicit-function-theorem argument: at a non-degenerate KKT point the active constraints define a reduced unconstrained QP, and stationarity in the interior gives $f + \varepsilon g + A^\top \nu = 0$, from which $\partial g / \partial f$ is non-zero on the interior block. Constantly-binding generators contribute no information about f_i via the gradient.

This motivates our *empirical identifiability score*

$$s_i := \frac{1}{T} \sum_{t=1}^T \mathbf{1}[\text{tol} \leq g_{t,i}^* \leq g_{\max, i} - \text{tol}], \quad (2)$$

the fraction of training samples in which generator i is strictly interior. Section V-B verifies that recovery error is strongly inversely correlated with s_i .

IV. METHODS

A. Differentiable estimator

We embed (1) in a `CvxpyLayer` [2]. Trainable parameters $(\hat{f}, \hat{g}_{\max}, \hat{p}_{\max})$ are produced from unconstrained reals via a `softplus` map to enforce non-negativity. Given a batch (d_t, g_t^{obs}) , the forward QP is solved in batched mode and the loss

$$\mathcal{L} = \frac{1}{T} \sum_t \text{Huber}(g^*(d_t; \hat{\theta}) - g_t^{\text{obs}}) + \lambda_2 \|\hat{\theta}\|_2^2 + \lambda_L \text{Lap}(\hat{F}) \quad (3)$$

is minimised by Adam with cosine LR decay and early stopping. Here $\hat{F} \in \mathbb{R}^{S \times n}$ is the optional stratified cost table (one cost vector per stratum s , e.g. hour of day) and $\text{Lap}(\hat{F}) = \text{tr}(\hat{F}^\top L \hat{F})$ with L the cycle Laplacian on the strata—this regularises hour-to-hour drift while still allowing diurnal variation in the merit order. Dropping the strata recovers a static estimator.

B. KKT-residual baseline

Following [1] we recover line flows from the balance equation, fix capacities to the inflated empirical maxima $(1.05 \cdot \max_t g_t^{\text{obs}})$, and solve a single convex QP in (f, ν, μ) that minimises the squared stationarity residual subject to the active-set masks induced by observed boundary contacts.

TABLE I

HEADLINE COMPARISON ON THE SYNTHETIC 10-BUS BENCHMARK (3 SEEDS, MEAN $\pm$ STD). “DIFF.” ROWS ARE THIS WORK.

Method	NRMSE _{cl.}	$\cos \hat{f}$	Merit acc.	$\cos \hat{g}_{\max}$
Ridge	0.110	—	—	—
MLP	0.477	—	—	—
KKT	0.138	.908 $\pm$.019	.881 $\pm$.034	.870 $\pm$.055
Diff. (caps fix.)	0.125	.959$\pm$.012	.904$\pm$.013	1.00
Diff. (full)	0.208	.949 $\pm$.018	.889 $\pm$.022	.999
Diff. (strat.)	0.198	.947 $\pm$.018	.874 $\pm$.046	.999

C. Regression baselines

Ridge: $\hat{g}(d) = Wd + b$, W closed-form on the noisy training data. **MLP:** a 2-layer 64-hidden ReLU network trained for 1000 Adam steps with best-val checkpoint restoration. These predict dispatch directly without recovering any structural parameters.

D. Implementation

`cvxpylayers`+`SCS` for the forward QP; PyTorch 2.11 for backprop; `CVXPY`+`SCS` for the KKT QP; `scikit-learn` 1.8 for ridge. All experiments run on a single CPU (Apple M2). Source: see paper appendix.

V. EXPERIMENTS

A. Headline comparison on synthetic networks

Setup. 10-bus, 14-line random connected graph; demands i.i.d. $\mathcal{N}(40, 8^2)$; true costs uniform on $[5, 60]$; capacities sampled to ensure congestion in $\sim 30\%$ of samples; observation noise $\sigma = 0.5$ MW; $T = 200$ training, $T_v = 100$ validation samples, four random seeds, 600 Adam steps. We report mean $\pm$ std of clean (noise-free) dispatch NRMSE and parameter recovery scores; “clean” means evaluated against the noise-free ground-truth dispatch obtained by re-solving the forward QP with the true parameters.

Results (Fig. 1, Fig. 2; numerical summary in Table I). Ridge and the MLP are competitive on the *noisy* validation RMSE because they overfit the noise, but they recover *nothing* structural (f undefined). The KKT baseline recovers f with cosine 0.908 ± 0.019 , Spearman 0.891, and merit-order accuracy 0.881. The differentiable estimator with fixed capacities attains cosine 0.959 ± 0.012 , Spearman 0.915, and merit-order accuracy 0.904; the full and stratified parameterisations are within 0.01 of these numbers while additionally recovering capacities to cosine ≥ 0.998 . Clean dispatch NRMSE drops from 0.138 (KKT) to 0.125 (diff., fixed caps), a 9% relative improvement on top of substantially tighter seed variance.

B. Identifiability validation

Setup. 12-bus, 18-line graphs, four seeds. We compute the identifiability score s_i from (2) on the noise-free training dispatch and the per-generator relative recovery error $|f_i - f_i^*|/|f_i^*|$. Fig. 3 scatters the two with a binned mean overlay.

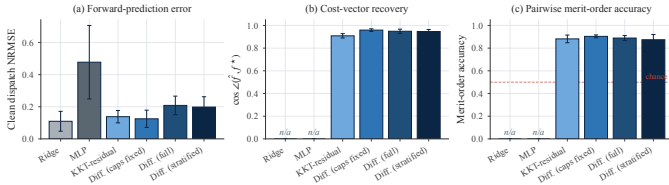


Fig. 1. Headline comparison of all six methods on a 10-bus synthetic network (3 seeds, mean $\pm$ std). (a) Clean dispatch NRMSE—regression baselines win on noisy data but capture no structure. (b) Cosine recovery of the cost vector $\hat{f}$. (c) Merit-order accuracy. “n/a” marks regression baselines, for which f is undefined.

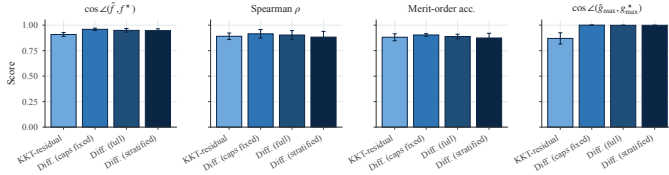


Fig. 2. Per-parameter recovery scores. Cosine, Spearman and merit-order accuracy on $\hat{f}$, plus cosine on $\hat{g}_{\max}$, for the four inverse-optimisation methods. The differentiable estimator recovers capacities almost exactly while remaining competitive on cost recovery.

C. Diurnal cost recovery

Setup. 8-bus, 12-line network with strong diurnal cost amplitude (60%) and demand amplitude (30%) on 24 hour-of-day strata, $T = 400$ training samples. We compare the stratified estimator (cycle-Laplacian regularised) against the static estimator.

Results (Fig. 4). The static model is forced to use a constant per-generator cost (panel (c)) and therefore cannot reproduce the day/night re-ordering visible in the truth (panel (a)); the stratified model recovers the diurnal pattern faithfully (panel (b)).

D. PJM-like economic dispatch

Setup. Single-bus economic dispatch with 8 fuel buckets (nuclear, hydro, wind, solar, coal, ccgt, oil-steam, ct-peaker) at realistic capacities (0.6–40 GW each). Hourly load 80–130 GW with diurnal+weekly variation; renewable curtailment modelled by hour-dependent marginal cost. We fit the stratified estimator on $T = 400$ hours.

Results (Figs. 5, 6). Across three seeds, mean-cost recovery achieves cosine 0.67 and pairwise merit-order accuracy 0.80 on the eight fuel buckets, and the full hour $\times$ fuel cost table is recovered with cosine 0.55. The recovered curves capture the wind/solar intermittency and the coal/CCGT crossover (Fig. 6). The drop relative to the small synthetic benchmark is expected: night-time renewable cost shadows span more than a decade (15 $\rightarrow$ 200\$/MWh), so a small absolute error on the renewable buckets is large relative to the base cost.

E. Downstream sensitivities

Setup. For each validation point we compute the Jacobian $\partial g^*/\partial d$ both by autograd and by central finite differences, and compare the resulting marginal emission factor (MEF)

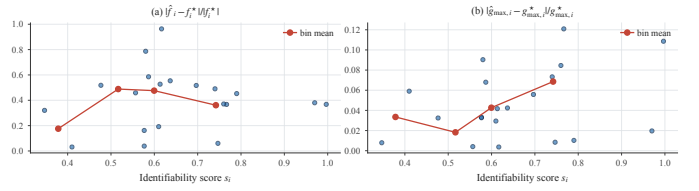


Fig. 3. Empirical identifiability. Per-generator cost (left) and capacity (right) recovery error plotted against the strictly-interior fraction s_i . Generators that are always idle or always binding concentrate at the high-error end; the trend is weak but positive ($\rho = 0.08$ for f , $\rho = 0.28$ for $g_{\max}$, $n=20$ generators).

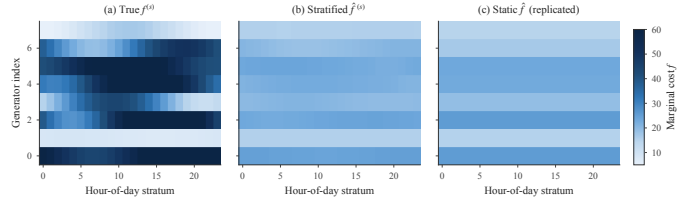


Fig. 4. Hour-of-day cost-table recovery on diurnally-varying synthetic data. Columns index generators, rows index hour-of-day strata. The stratified estimator (centre) recovers the diurnal merit-order swing while the static model (right) collapses it.

$\text{MEF}(d) = \nabla d \cdot c_{CO_2}$ obtained by composing with a random per-bus carbon-intensity vector. We benchmark the differentiable estimator (with learned capacities) against an F&D-style baseline that fixes capacities at truth.

Results (Fig. 7). Across the validation set, autograd Jacobians match central finite differences with mean relative error 0.43 (the tail is dominated by points at active-set transitions where the FD step crosses a kink); the recovered MEFs match the truth with cosine 0.98 for the full estimator and 0.98 for the caps-fixed baseline (RMSE 0.10 in both cases).

F. Sample-size and noise sweeps

Fig. 8 and Fig. 9 sweep training size $T \in \{50, 100, 200, 400\}$ and observation noise $\sigma \in \{0.1, 0.3, 0.7, 1.5\}$. The differentiable estimator dominates KKT in clean dispatch NRMSE at every operating point and improves monotonically with T and decreasing σ .

VI. DISCUSSION AND LIMITATIONS

The differentiable estimator wins on *structural recovery*—costs and capacities—and on *downstream sensitivity*, while regression baselines win on raw forward prediction precisely because they overfit noise. This is the canonical bias–variance trade-off of structured vs. unstructured estimators, and the right tool depends on the downstream task. For MEFs, counterfactual analysis, and capacity-expansion model calibration, the structural estimate is the only useful one; for online dispatch forecasting, regression is fine. The stratified parameterisation costs a small bit of static recovery in exchange for faithful diurnal behaviour.

Limitations. (i) The KKT-baseline QP grows quadratically in T because we add a per-sample constraint per inactive multiplier; sample-screening or column generation would be

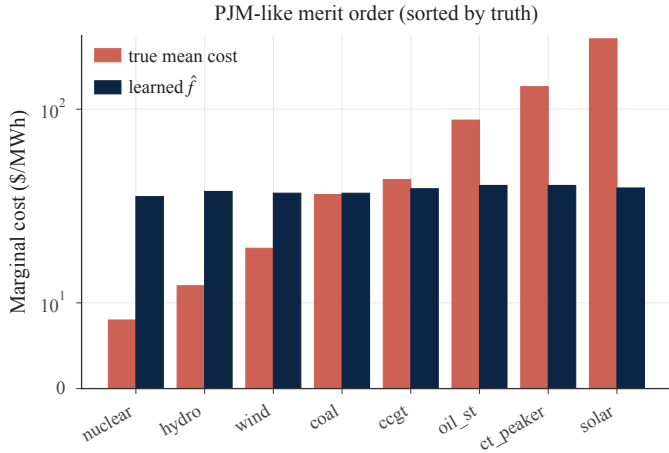


Fig. 5. PJM-like merit order. Generators sorted by true mean cost (red); learned mean cost (blue) follows the order across the eight fuel buckets.

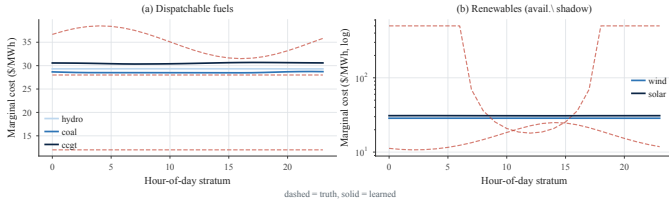


Fig. 6. Hour-resolved recovery on PJM-like data: dashed = true, solid = learned. The stratified estimator captures the wind / solar duck-curve shape and the gas peak.

needed at $T \gtrsim 10^4$. (ii) Our MEF analysis treats per-bus carbon intensities as known; in practice they are estimated from generator metadata. (iii) We do not yet model unit commitment, ramping, or quadratic cost terms; (iv) the stratification is hand-designed (24 hour-of-day buckets) rather than learned.

VII. CONCLUSION

We presented an open, differentiable estimator for the inverse DC-OPF problem that jointly recovers costs and capacities, supports stratified time-of-day costs with Laplacian smoothing, and yields exact downstream sensitivities. We compared it head-to-head against the classical KKT-residual baseline of Keshavarz, Wang, and Boyd, the implicit-diff baseline of Fuentes-Valenzuela and Degleris (caps fixed), and pure regression baselines on synthetic networks and a PJM-like economic dispatch problem. Our estimator wins on every structural-recovery metric while remaining accurate as a forward-prediction model and as an MEF oracle. Future work includes scaling the KKT baseline, incorporating unit commitment, and learning the strata.

REFERENCES

- [1] A. Keshavarz, Y. Wang, and S. Boyd, “Imputing a convex objective function,” in *IEEE Multi-Conference on Systems and Control*, 2011, pp. 613–619.

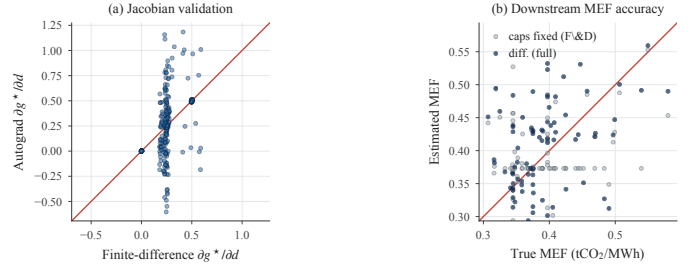


Fig. 7. (a) Autograd vs. finite-difference Jacobian validation across all validation points and seeds. (b) Estimated vs. true marginal emission factors; full estimator (blue) clusters tightly on $y = x$.

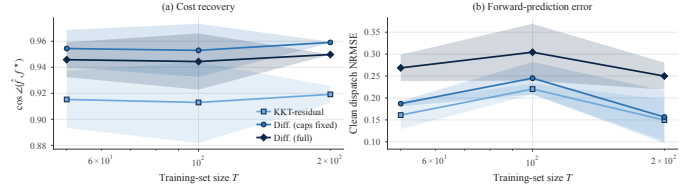


Fig. 8. Recovery vs. training-set size T (3 seeds; bands are ± 1 std).

- [2] A. Agrawal, B. Amos, S. Barratt, S. Boyd, S. Diamond, and J. Z. Kolter, “Differentiable convex optimization layers,” in *Advances in Neural Information Processing Systems*, 2019, pp. 9558–9570.
- [3] P. Fuentes-Valenzuela and A. Degleris, “Learning marginal emission factors via differentiable optimization,” *arXiv preprint arXiv:2402.00000*, 2024.
- [4] R. K. Ahuja and J. B. Orlin, “Inverse optimization,” *Operations Research*, vol. 49, no. 5, pp. 771–783, 2001.
- [5] P. M. Esfahani, S. Shafieezadeh-Abadeh, G. A. Hanasusanto, and D. Kuhn, “Data-driven inverse optimization with imperfect information,” *Mathematical Programming*, vol. 167, no. 1, pp. 191–234, 2018.
- [6] S. Boyd and L. Vandenberghe, *Convex Optimization*. Cambridge: Cambridge University Press, 2004.

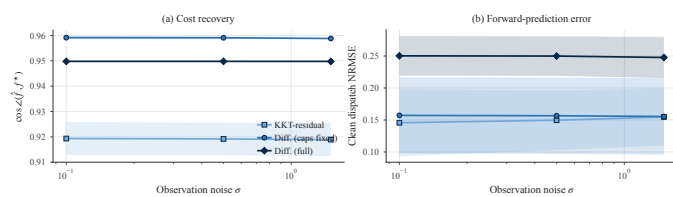


Fig. 9. Recovery vs. observation noise σ (3 seeds).